Is This Problem Worth Two Semesters of Your Life?

Unit I · Lecture I.4 — Novelty, feasibility, relevance, measurability, ethical soundness: the FINER test

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What you should be able to do by 55 minutes from now

LO1 · Name
State the five FINER criteria and the sixth that computer science adds, and say what each one is protecting you from.

LO2 · Score
Apply a five-point rubric to any research problem — including a stranger's — and defend each score with evidence.

LO3 · Diagnose
Given a failing problem statement, identify *which* criterion failed rather than saying "it's bad".

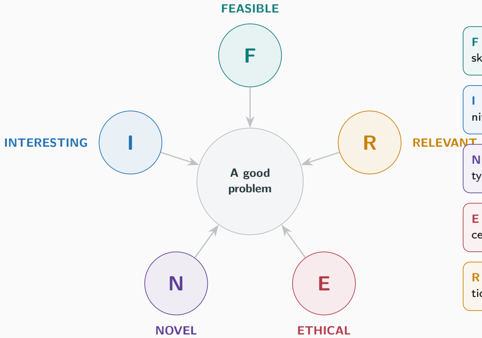
LO4 · Repair
Apply the standard repair move for each failed criterion, and know when the honest move is to abandon the problem.

Maps to CO1 *Identify and formulate research problems using defined criteria and characteristics.*

Carry-in: I.1 gave you the four ingredients. I.2 gave you five sources. I.3 gave you the narrowing funnel and a written statement. Today: the test that statement has to pass.

FINER · the five-criteria filter

FINER — borrowed from clinical research, and it travels well



F · Can you actually do it? Data, compute, time, skills, access, permission.

I · Does anyone care? You, your advisor, a community, an industry.

N · Does it add something? Not merely "nobody has typed this exact sentence".

E · Should it be done? Consent, harm, dual use, licences, privacy.

R · What changes if you succeed? Knowledge, practice, policy or cost.

Hulley, Cummings, Browner et al., *Designing Clinical Research*, 1988 — and still the default checklist on grant panels today.

F · Feasible — the multiplication test

DATA — can you get it, legally? → **COMPUTE** — GPU-hours you own → **TIME** — weeks you truly have → **SKILL** — what you can learn in 4 weeks → **ACCESS** — people, labs, permission

If any one of these is zero, the product is zero.

The optimism you are about to display

Students estimate the *coding* time and forget the other four. Coding is usually **under a third** of the total.

The one-week rule

If you cannot produce a **crude end-to-end result in one week** — bad numbers, tiny sample, ugly code — the problem is probably not feasible at your scale.

I · Interesting — interesting *to whom*?

Only you → **Your advisor** → **A lab / group** → **A research community** → **An industry or a public**

A hobby.
Perfectly fine —
not a thesis.

Minimum bar
for a final-year
research project.

Where papers get
cited and startups
get funded.

"Who would be annoyed if this question stayed unanswered?"

The test that cannot be faked

Name **one specific person or group** — not "society", not "researchers" — who would read your result and change what they do.

N · Novel — four kinds, and only one of them is "nobody did it"

1 · New answer
A question already asked, answered better.
Faster, cheaper, more accurate, smaller.

2 · New question
Nobody thought to ask it.
Rarest and riskiest. Usually not a student project.

3 · New context
Known method, untested setting.
Indic languages, low-end hardware, rural networks.

4 · New evidence
A claim everyone believes, actually tested.
Replication, negative results, ablations.

Two false beliefs about novelty

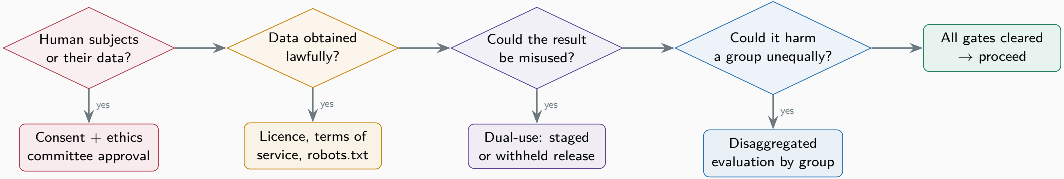
(a) "If a paper exists, the topic is dead."
(b) "If no paper exists, the topic is good."

Both are wrong. (b) is more dangerous — sometimes nobody asked because the answer is obvious or useless.

Where your novelty will realistically come from

Columns **3** and **4**. A careful evaluation of a known method in a context nobody has tested is a publishable, defensible, finishable final-year contribution.

E · Ethical — five gates, and any one can stop you



"Publicly visible" is not the same as "public"; "public" is not the same as "yours to use". Unit III returns to all of this in depth.

R · Relevant — the "so what" chain

Your result → so what? → so what? → someone acts differently

Worked chain. "Our detector cuts false alerts by 40%" → *so what?* → "an analyst triages 60 alerts a shift instead of 100" → *so what?* → "the real intrusion in the queue gets looked at on day one instead of day four."

Relevance has four legal currencies

Knowledge (we understand something we did not) · **Practice** (someone builds differently) · **Policy** (a rule changes) · **Cost** (the same outcome, cheaper).

Relevance theatre

"This will help society and contribute to the nation's growth."

A sentence that fits every project fits none. If your relevance claim would survive being pasted into someone else's proposal, it is not a relevance claim.

Your turn · scoring a real problem

Think–Pair–Share · score this statement out of 25

Submitted by a final-year team, lightly edited

"We propose to build a deep learning model that predicts student dropout at our university using data from the student information system, and to show that our model outperforms existing approaches, thereby helping the institution improve student outcomes."

Think 2 min · **Pair** 3 min · **Share** 3 min

Alone (2 min). Score each of F, I, N, E, R from 1 to 5. Write *one word of evidence* beside each score.

In pairs (3 min). Find the criterion where you disagree most. Argue it out.

Share (3 min). I want the **lowest** score in the room and the reason — not the average.

Scoring key

5 — clearly satisfied, evidence stated
4 — satisfied, evidence implied
3 — arguable either way
2 — probably fails, repairable
1 — fails; problem must change

Any single 1 is fatal. You do not average your way past a 1.

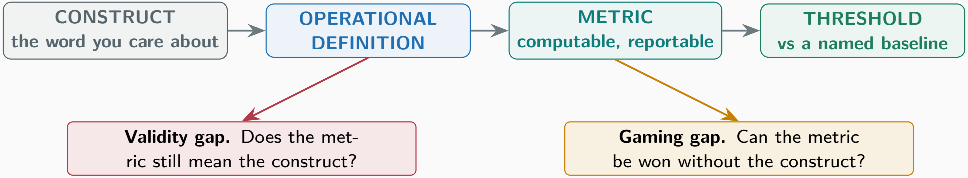
The FINER scorecard — print this, use it on every idea

Criterion	The question you must answer with evidence	Score 1 looks like
F	Do I have the data, compute, weeks, skills and permission — *today*, not in principle?	The dataset requires an approval I have not applied for.
I	Can I name a specific person or group who would change what they do?	The only honest audience is "me".
N	Which of the four novelty types is mine, and which three papers am I standing next to?	The exact study exists, with more data than I have.
E	Whose data, whose risk, who could be harmed, and who approved it?	Personal data, no consent, no approval route.
R	What decision or belief changes if my result is true? And if it is false?	Nothing changes either way.
M	**(CS addition)** What exactly do I measure, against what baseline, and what number counts as success?	"Better performance" with no metric named.

Six rows, thirty points. Below 20, or any single 1 — do not start.

The sixth criterion computer science adds · Measurability

M · Measurable — from a word you like to a number you can defend



Construct	Operational definition	Metric	Threshold
"code quality"	defects surviving review	defects / KLOC	$\leq$ baseline $-$ 20%
"usable"	task completed unaided	completion rate, time-on-task	$\geq 85\%$, $\leq 90\,\mathrm{s}$
"efficient"	inference on the target device	p95 latency, peak RAM	$< 100\,\mathrm{ms}$, $< 512\,\mathrm{MB}$
"fair"	error parity across groups	max group FPR gap	$\leq$ 5 percentage points

Four real projects, scored the way a reviewer would

Project	F	I	N	E	R	M	The interesting tension
AlphaGo (DeepMind, 2016)	1	5	5	5	4	5	Infeasible for anyone else on Earth. Feasibility is *relative to you*, not absolute.
GitHub Copilot productivity study	4	5	3	3	5	2	"Productivity" is the hardest construct in software engineering. Self-reported speed-up is not speed-up.
UPI (NPCI, India, 2016–)	3	5	4	4	5	5	Relevance at national scale; novelty came from *system design*, not from an algorithm.
IBM Watson for Oncology	2	5	4	2	5	1	Trained largely on synthetic cases; "agreement with our own oncologists" is not an outcome measure.

Scores are the instructor's, for argument. Disagree with them — that is the exercise.

Stress-testing · spot the failure

Spot the mistake · which single criterion fails hardest?

A "We will use an LSTM to predict next-day stock prices with 99% accuracy and build an automated trading system."

B "We will scrape two million public Instagram profiles of teenagers to detect early signs of depression from their captions."

C "We will compare bubble sort, insertion sort and quicksort on randomly generated arrays and report which is fastest."

D "We will improve the user experience of the university website using modern web technologies."

E "We will apply a transformer architecture to our college's canteen billing data."

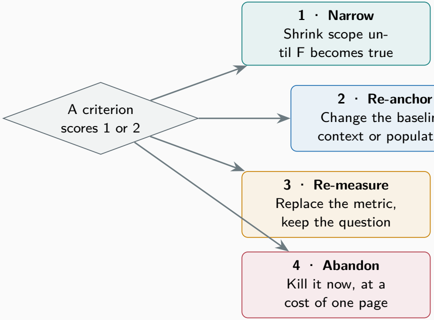
Two minutes, in threes

For each statement write **one letter** — F, I, N, E, R or M — and nothing else. No explanations yet.

Spot the mistake · answers, and the repair

	Fails	Why	The repair
A	**F** (then N)	99% next-day accuracy contradicts decades of efficient-market evidence. If it worked you would not be publishing it.	Change the question: does sentiment add *anything* over a random-walk baseline? Report an honest, small effect.
B	**E**	Minors, sensitive health inference, no consent, terms-of-service breach. Visible is not consented.	Use an existing consented, ethics-approved corpus. Or study the detection method on adult, opt-in data.
C	**N** (and R)	Settled since the 1960s. Absence of recent papers means closed, not open.	Move to a context nobody has measured: sorting on a specific embedded cache hierarchy, or energy per element.
D	**M**	"Improve user experience" names no construct, no metric, no baseline. Unfalsifiable.	Task completion rate and time-on-task for three named tasks, measured before and after, $n \geq 30$.
E	**R** (then I)	Method-first, not problem-first. Nothing changes for anyone if it works.	Start from a person's pain: can the canteen cut the 12:30 queue? Then choose whatever method fits — probably not a transformer.

When a criterion fails, there are only four moves



Fixes **F** almost always. The default move, and the one students resist.

Fixes **N** and **I**. Slide 8, column 3 — new context is real novelty.

Fixes **M** and often **R**. Cheapest repair on this slide.

The only move for a hard **E** failure. Not a defeat — a saved semester.

The move that is not on this list

"Keep the problem and hope the criterion stops mattering." It does not stop mattering. It surfaces at your review panel, in front of an audience.

Feasibility is arithmetic — run it before you commit

```python
# FINER, the F row: the number your optimism does not want you to compute.
WEEKS_LEFT = 14 # be honest: subtract exams, placements, festivals
HOURS_PER_WEEK = 12 # per team member, sustainable, not heroic
MEMBERS = 3
GPU_HOURS_OWNED = 60 # what you can actually book, not what exists

budget_h = WEEKS_LEFT * HOURS_PER_WEEK * MEMBERS

plan = { # your honest estimate, in person-hours
    "get + clean data" : 70, # students guess 20; it is never 20
    "reproduce a baseline" : 45, # if you cannot, you have no comparison
    "build your method" : 60,
    "run experiments" : 40,
    "ablations + repeats" : 35, # 3 seeds minimum, or your result is noise
    "write-up + figures" : 55,
}
spent = sum(plan.values())

print(f"budget {budget_h} h | plan {spent} h | slack {budget_h - spent} h")

runs = 4 * 3 # configurations x random seeds
gpu_h = runs * 3.5 # measured on ONE short pilot run, never guessed
assert gpu_h <= GPU_HOURS_OWNED, f"need {gpu_h} GPU-h, have {GPU_HOURS_OWNED}"
assert spent <= 0.7 * budget_h, "no slack: one bad week ends the project"
```

Two assertions. If either fails, you have a scoping decision to make — today, not in November.

Sixty-second debate · is novelty overrated?

Left half of the room argues

“Novelty first.”

Science advances by new things. Conferences reject the incremental. Careers are built on being first. A replication has never founded a company.

Right half of the room argues

“Evidence first.”

A field where nothing is checked accumulates false results. Many published gains vanish against properly tuned baselines. Verified knowledge beats novel noise.

Thirty seconds each side, then I want the synthesis

Which criterion is each side implicitly ranking above the other?

Guided practice and checkpoint

Guided practice · score the statement in your hand

Four minutes, alone, then swap

1. Score your own I.3 statement on all six rows. Write one piece of *evidence* beside each score — a file you have downloaded, a paper you have read, a person you have named.

2. Circle your **lowest** row. That is your project's real risk, whatever you were worrying about instead.

3. Choose one of the four repair moves and rewrite *one sentence* of your statement.

4. Swap with a neighbour. Score theirs cold. Compare.

What almost always happens

Your neighbour scores you **lower than you scored yourself** — most often on **M** and **R**.

They are usually right. You have context in your head that is not in the sentence, and a reviewer only ever gets the sentence.

If you scored a 1

Come and talk to me this week. A 1 in week three is cheap. A 1 discovered in week twelve is a semester.

Checkpoint quiz · write all four, then we reveal

Q1 · one word

A problem scores 5 on every criterion except one, where it scores 1. Should you start? Answer in one word, then one sentence of justification.

Q2 · name the type

"We evaluate three well-known summarisation models on Kashmiri and Dogri news text, for which no benchmark exists." Which of the four novelty types is this, and is it enough?

Q3 · two gaps

A team measures "developer productivity" as commits per week. Name the validity gap and the gaming gap in one line each.

Q4 · choose the move

Your problem needs hospital records that require an approval taking eight months. Which of the four repair moves applies, and which one definitely does not?

Three minutes. No discussion, no phones. Ninety seconds of silence is part of the exercise.

Checkpoint quiz · answers

	Answer
Q1	**No.** The criteria are a product, not a sum — a single 1 zeroes it. A brilliant, important, novel, ethical problem you cannot execute produces nothing at all.
Q2	**Type 3, new context** — and yes, it is enough, *provided* M is answered. You must state the metric (ROUGE against what reference? human adequacy ratings by how many annotators?) and how the evaluation set was built. Without M this is a demo, not a study.
Q3	**Validity gap:** commits measure activity, not value — a developer who spends a week on a one-line fix to a critical bug scores near zero. **Gaming gap:** anyone who knows the metric can split one change into fifteen commits. The measure survives; the construct does not.
Q4	**Applies:** move 2, re-anchor — use a public de-identified corpus (MIMIC-style), or a synthetic or simulated cohort, and state the limitation honestly. **Definitely does not apply:** move 1, narrowing. A smaller slice of inaccessible data is still inaccessible. Narrowing fixes scope, never access.

Close

Reflection and key takeaways

Turn to your neighbour · thirty seconds each

1. Which criterion did you score *highest* before this lecture and *lowest* after?

2. What surprised you?

3. Name one place outside research — a job offer, a startup idea, a purchase — where you could run FINER tonight.

Five things to carry out of this room

1. The criteria multiply. **Any single 1 is fatal.**

2. **Feasibility is relative to you**, not absolute. AlphaGo scores 5 for DeepMind and 1 for you.

3. **Novelty is four things**, and yours will be *new context* or *new evidence*. That is respectable.

4. **Computer science adds M.** If you cannot name the metric, the baseline and the threshold, you do not yet have a problem.

5. **Ethics is a gate, not a form.** You cannot narrow your way past it.

CO1 **LO1–LO4 met** **Scorecard collected** **Next: I.5 — the errors people actually make**

Homework, reading and what comes next

Homework — due at the start of I.5

H1. Completed **FINERM scorecard** for your own statement: six scores, six pieces of evidence. One page.

H2. Run the **feasibility script** with honest numbers. Paste the output; if an assertion fails, name the repair move you chose.

H3. Score **one published paper** from your area on all six rows. It will be easier than scoring yourself — notice that.

H4. One sentence: your project's **single biggest risk**, named as a criterion.

Further reading

Textbook. Melville & Goddard, Ch. 2. Ganesan — problem selection criteria.

Origin. Hulley et al., *Designing Clinical Research*, Ch. 2 — four pages.

Read in full. Beede et al., *A Human-Centered Evaluation of a Deep Learning System Deployed in Clinics*, CHI 2020.

Skim. Ferrari Dacrema et al., *Are we really making much progress?*, RecSys 2019.

Optional. Wagstaff, *Machine Learning that Matters*, ICML 2012.

Bring to I.5

Your scorecard, and your lowest criterion. I.5 is a catalogue of the errors that produce each low score — yours will be in it.

**“Not everything that can be counted counts,
and not everything that counts can be counted.”**

— William Bruce Cameron, 1963

(routinely, and wrongly, attributed to Einstein)